Final Project Data Analyst

Zomato Delivery Performance Analysis : Factors Influencing Delivery Time

Introduction

Felicia Angjaya

Finance student with strong interest in data analytics & business insights

Education

Finance, BINUS University – Semester 5

linkedin.com/in/feliciaangjaya

Felangjaya@gmail.com

Overview Project

Zomato Delivery Performance Analysis: Factors Influencing Delivery Time

This project focuses on analyzing the key factors that influence Zomato's delivery time. External factors include traffic conditions, weather, and delivery distance, while internal factors cover driver age, vehicle condition, and multiple deliveries. The analysis shows that heavy traffic, bad weather, and longer distances increase the risk of delays, while poor vehicle conditions and multiple orders add more variance to delivery time.

Objectives

1. **Analyze key factors influencing delivery performance,** particularly delivery time
2. **Compare delivery times across different operational conditions** such as traffic density, weather, and vehicle condition
3. **Provide actionable insights** to improve operational efficiency and customer satisfaction

Goal

To enhance delivery efficiency and service consistency by identifying the key factors that influence delivery time, enabling data-driven decisions to reduce delays, improve on-time delivery rates, and maintain high customer satisfaction

Tools

Power BI

Streamlit

Project Background

In the highly competitive food delivery industry, delivery time is one of the most critical factors influencing customer satisfaction and loyalty. Even small delays can impact customer experience, restaurant operations, and overall service quality. With rising customer expectations and increasing competition, companies like Zomato must identify and address the key factors that affect delivery time to maintain efficiency and competitiveness.

Project Overview

This project evaluates Zomato's delivery performance using historical operational data on drivers, locations, environmental and vehicle conditions, and orders. ***Early findings reveal notable delivery time variations for similar distances, suggesting inefficiencies in route planning, driver allocation, or operations.*** The analysis aims to identify the drivers of these inconsistencies and their impact on customer experience and business results

Why This Project Matters?

Consistent delivery time is crucial for customer satisfaction, efficiency, and competitiveness. Understanding the internal (driver skill, vehicle condition, workload) and external (traffic, weather, Vehicle Condition) factors behind these variations enables Zomato to implement targeted improvements and strengthen on-time performance

Expected Outcomes

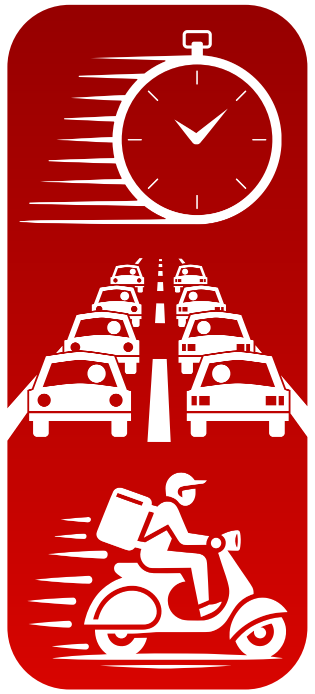
1. **Clear identification of factors with the greatest impact on delivery time**
2. **Comparative analysis of delivery time** under different operational conditions (traffic, weather, vehicle condition)
3. **Actionable insights** to improve operational efficiency and customer satisfaction

Beneficiaries

- The **operations team** can optimize driver allocation and delivery routes
- **Management** can make well-informed operational decisions
- **Restaurant** partners benefit from more dependable delivery services
- **Customers** enjoy faster and more consistent deliveries

Business Problem

Problem Statement



Zomato experiences inconsistent delivery times that may reduce customer satisfaction and efficiency

Factors like traffic, weather, and vehicle condition may cause delays, but their exact impact is unclear

Fluctuating delivery performance makes it hard to maintain service quality and competitiveness

Why This Problem Matters

1. **Delivery Time Drives Business Success**
   Delivery time is a critical factor in the food delivery industry, directly influencing customer satisfaction, loyalty, and brand reputation

2. **Lack of Clarity on Delay Causes**
   Unclear causes of delays prevent Zomato from implementing targeted operational improvements to address the root issues

3. **Operational and Competitive Gains from Resolution**
   Resolving this problem will improve operational efficiency, increase on-time delivery rates, and strengthen Zomato’s market competitiveness

Business Impact – If the Problem Is Resolved

Data Understanding & Preprocessing

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Dataset from **Kaggle**

Zomato Delivery Operations Analytics Dataset
Exploring Delivery Dynamics and Customer Experience on Zomato
kaggle.com

45584 Number of Records

20 Number of Variables

9 Numeric

11 Categorical

The Zomato Delivery Dataset contains historical operational data, including driver information, ratings, locations, order times, distances, and environmental conditions

Number of duplicate rows : 0 Duplicates

Note : *All other columns have 0 missing values*

Column	Missing Values
Delivery_person_Age	1854
Delivery_person_Ratings	1908
Time_Orderd	1731
Weather_conditions	616
Road_traffic_density	601
multiple_deliveries	993
Festival	228
City	1200

Handling Missing Values (dropna method)

Step	Number of Records	Missing Values
Before Cleaning	45,584	Yes
After Removing	41,359	No

Removed **~9.27%** of data

Data Type Conversion

- **Order_Date** — Converted to **datetime** format for accurate date-based analysis
- **Time_Orderd** — Converted to **time** format for better time-based calculations
- **Time_Order_picked** — Converted to **time** format for better time-based calculations

No invalid values were found in the dataset

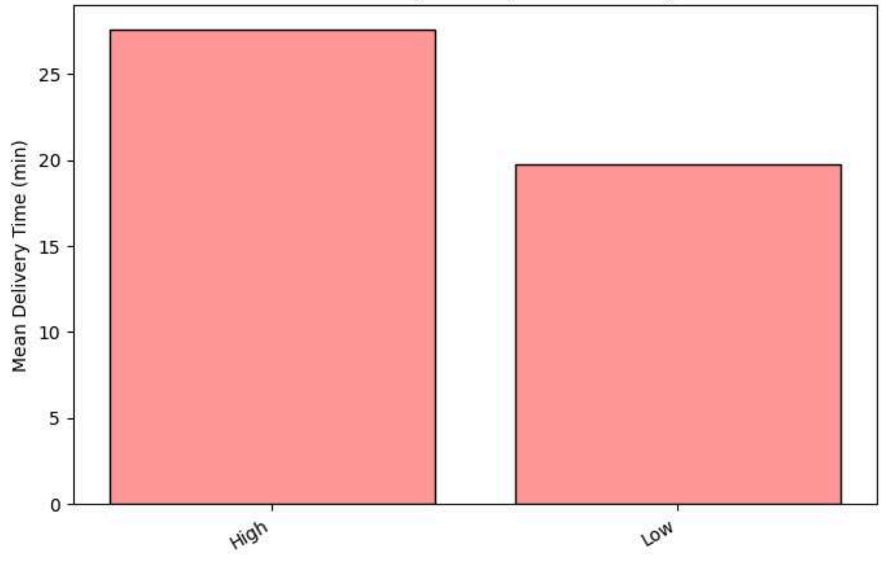
Data Analysis & Insights

Technique Used The analysis applied **descriptive** and **exploratory data analysis (EDA)** techniques to uncover patterns in delivery performance

External Factors Affecting Delivery Time on Zomato

Based on the predominant delivery distance of 1.5 km (using 1-2km)

(Mean) Delivery Time by Traffic Density

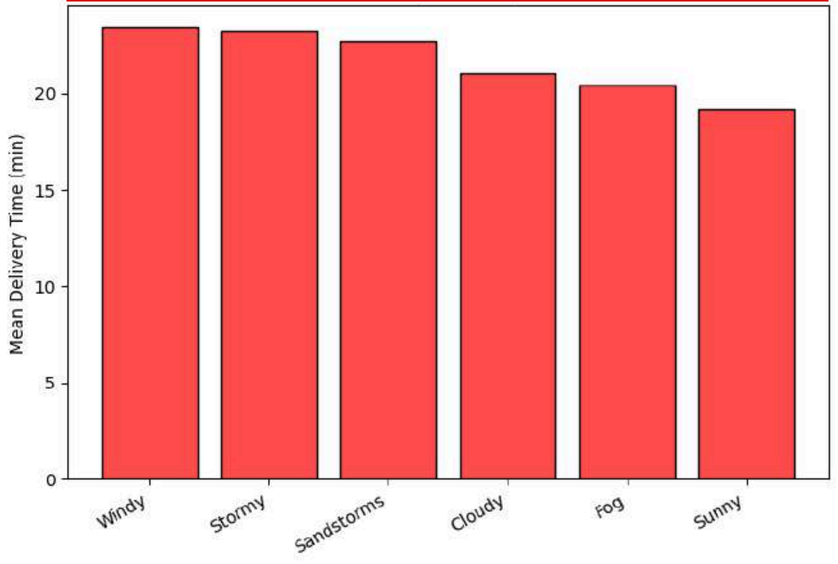


Mean Delivery Time by Traffic Density

road_traffic_density	mean_time
High	27.610262
Low	19.728359

Deliveries in **high traffic areas take around 27.6 minutes** on average, which is **significantly slower** compared to 19.7 minutes in low traffic, indicating congestion plays a major role in delays

(Mean) Delivery Time by Weather

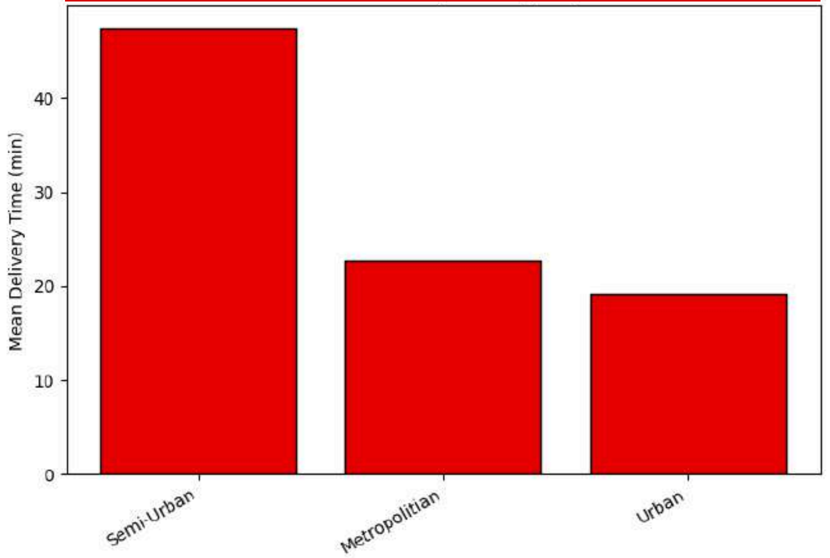


Mean Delivery Time by Weather

weather_conditions	mean_time	median_time
Windy	23.392361	24.0
Stormy	23.242284	24.0
Sandstorms	22.734597	23.0
Cloudy	21.034258	19.0
Fog	20.432343	19.0
Sunny	19.232945	18.0

Windy and stormy weather extend delivery times to about 23.3 minutes, whereas **sunny conditions are faster** at 19.2 minutes, suggesting adverse weather disrupts delivery efficiency

(Mean) Delivery Time by City

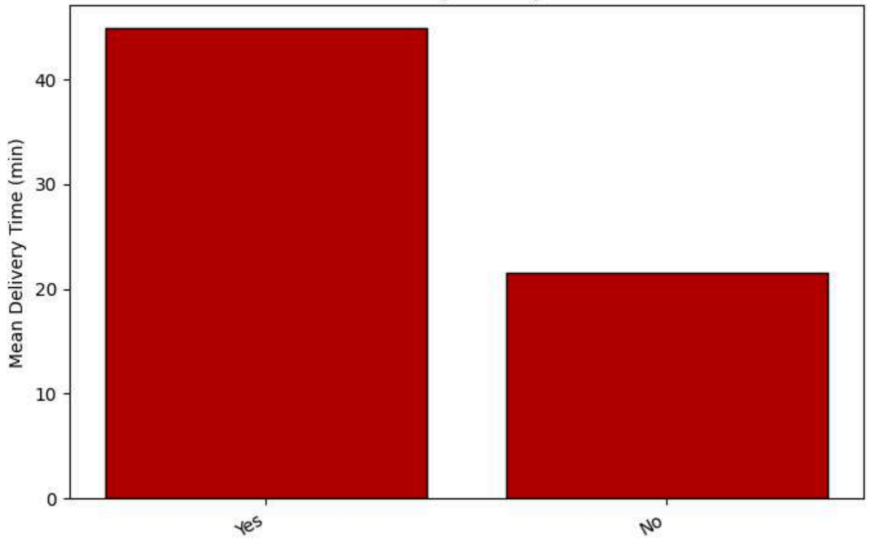


Mean Delivery Time by City

city	mean_time	median_time
Semi-Urban	47.500000	47.5
Metropolitian	22.671012	22.0
Urban	19.250234	18.0

Semi-urban areas show extremely long delivery times, averaging 47.5 minutes, compared to 19.3 minutes in urban areas, likely due to infrastructure and accessibility challenges

(Mean) Delivery Time by Festival



Mean Delivery Time by Festival

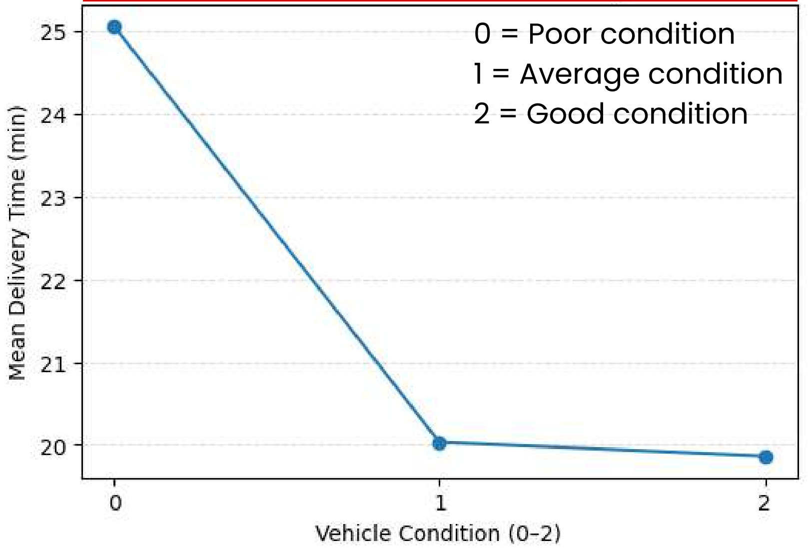
festival	mean_time	median_time
Yes	44.833333	44.5
No	21.578027	21.0

During **festivals**, **delivery times spike to an average of 44.8 minutes, more than double** the 21.6 minutes **on normal days**, indicating that seasonal demand significantly slows down delivery operations

Data Analysis & Insights

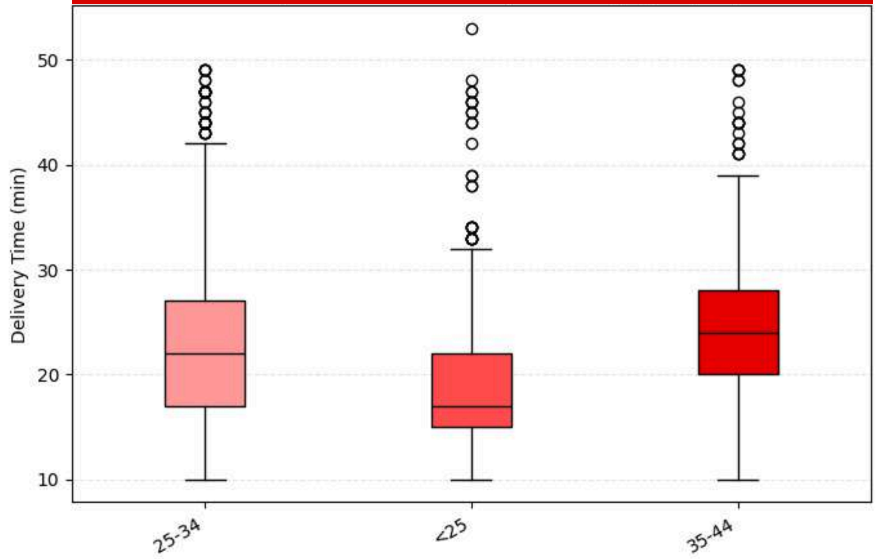
Internal Factors Affecting Delivery Time on Zomato

Vehicle Condition vs Delivery Time



Poorly maintained vehicles (condition 0) take the longest to deliver at ~25 minutes on average, whereas well-maintained vehicles (condition 2) reduce delivery time to under 20 minutes, showing the operational impact of maintenance

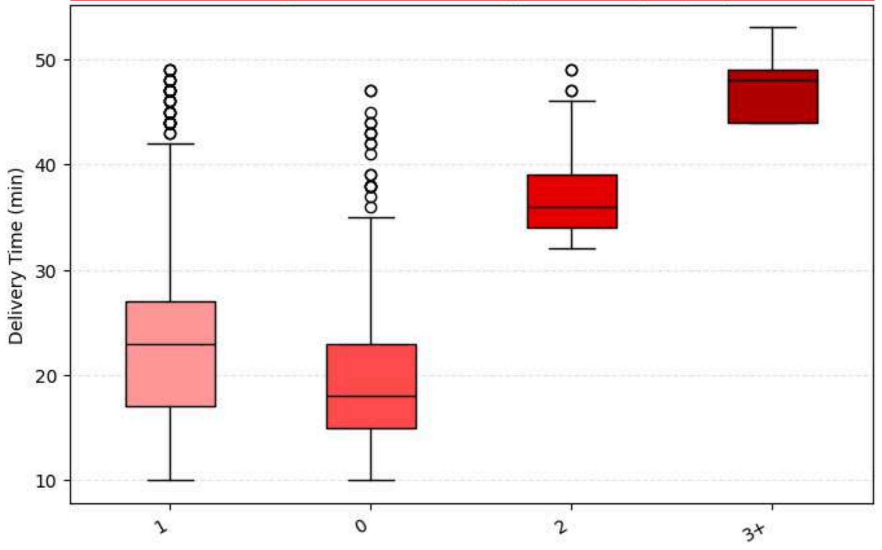
Delivery Time Distribution by Driver Age



Younger drivers (<25) tend to deliver slightly faster, with a lower median time and fewer extreme delays, while older age groups (35–44) show higher median times and more variability

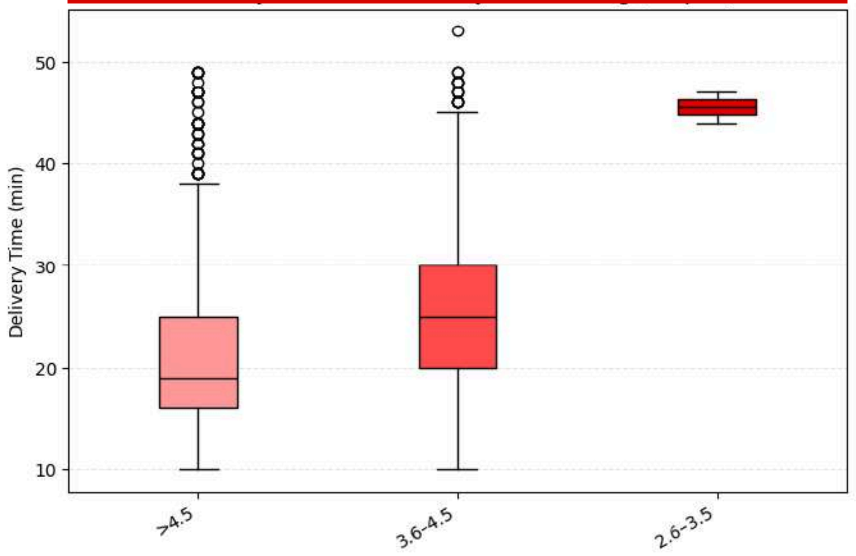
High workloads (3+ orders) cause a dramatic slowdown, pushing median delivery times close to 50 minutes, compared to around 17–25 minutes when handling one or zero orders

Delivery Time Distribution by Workload



Drivers with ratings above 4.5 complete deliveries faster and more consistently, while those rated 2.6–3.5 have both the slowest delivery times (median ~46 minutes) and the least variability, possibly due to consistently poor performance

Delivery Time Distribution by Driver Rating



**Internal factors are variables originating within the organization's operations that can be directly managed or influenced, while external factors are variables arising from the surrounding environment that are beyond the organization's direct control*

Data Analysis & Insights

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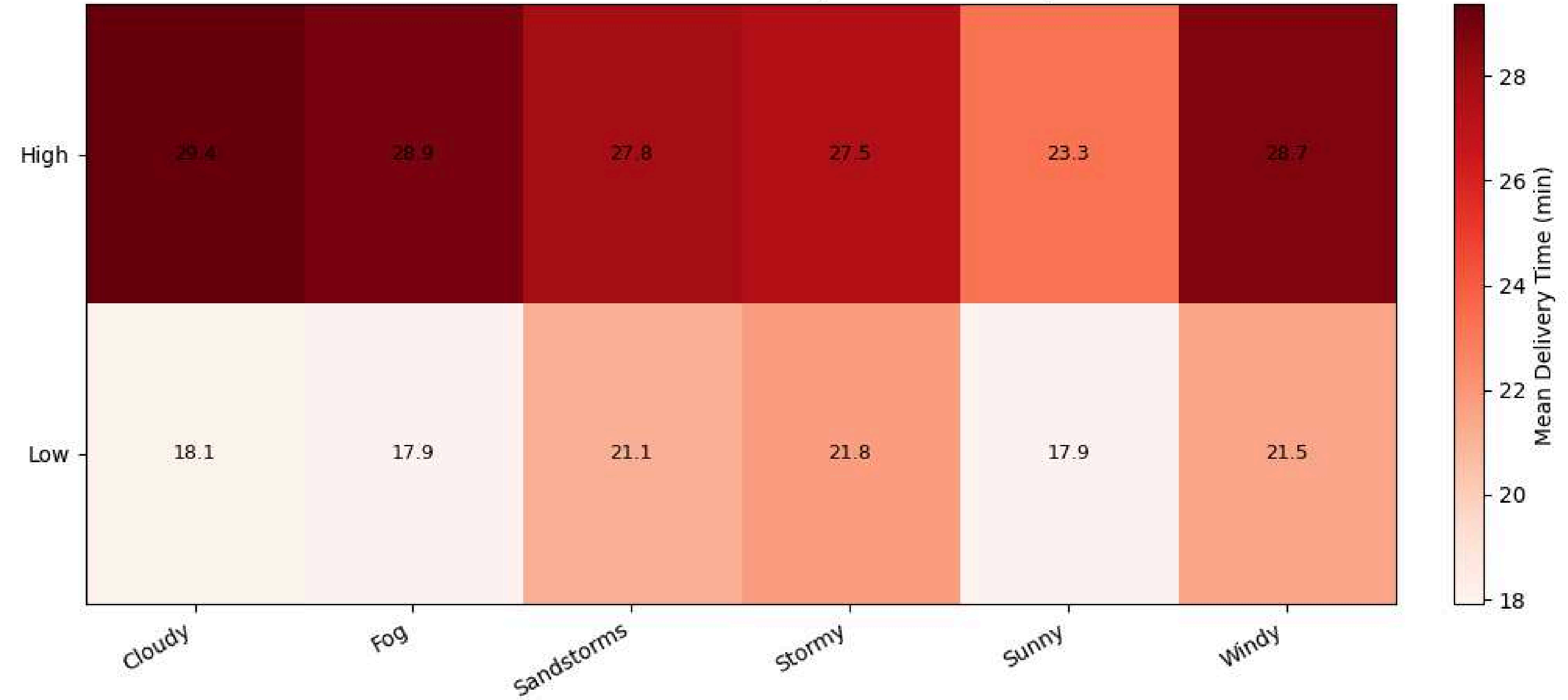
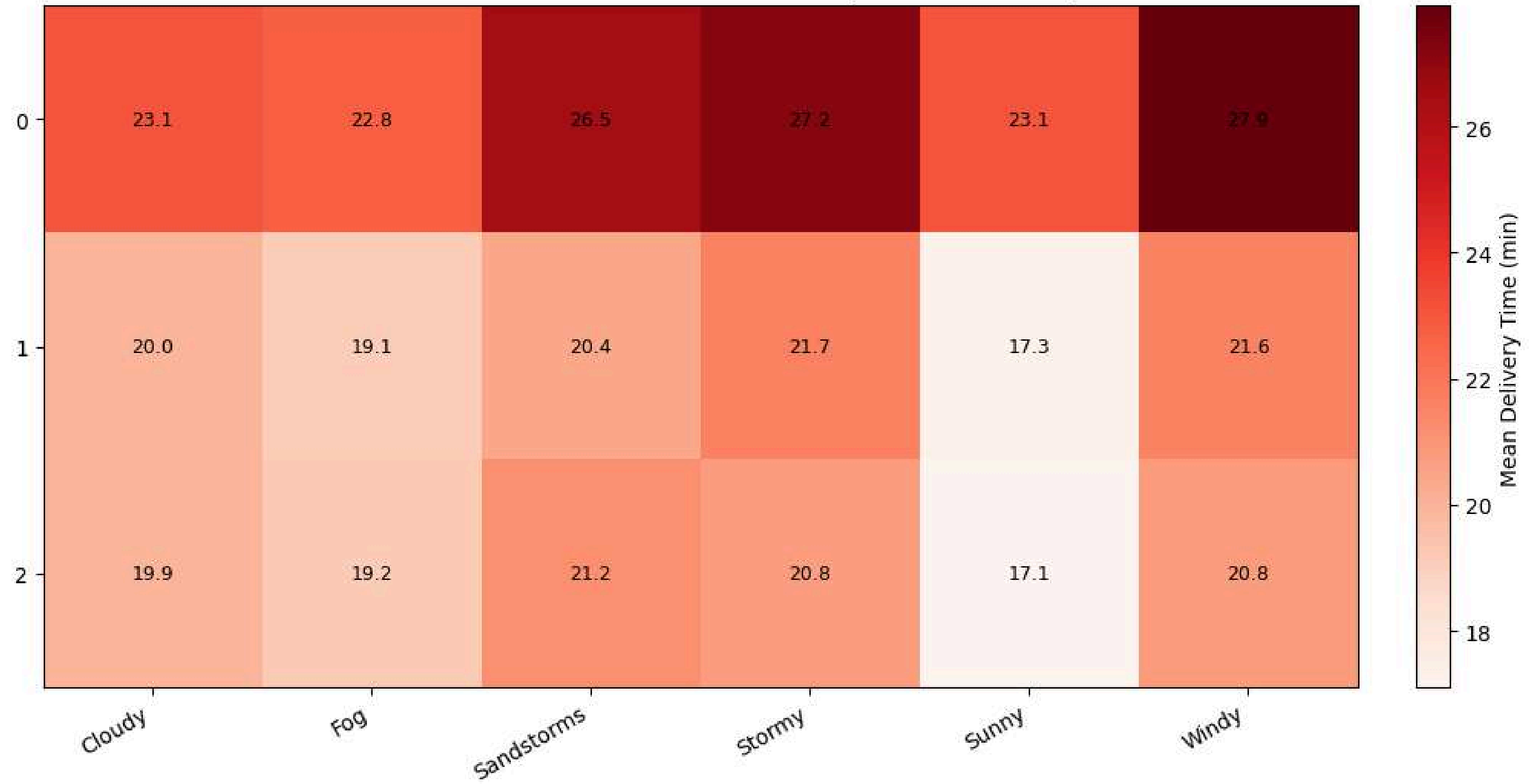
Interaction analysis

Interaction : Traffic x Weather



Interaction : Vehicle Condition x Weather



High traffic combined with cloudy, foggy, or windy conditions results in the longest delivery times, reaching around 29 minutes

Low traffic consistently leads to faster deliveries across all weather types, indicating that traffic is a more dominant factor than weather in slowing down deliveries

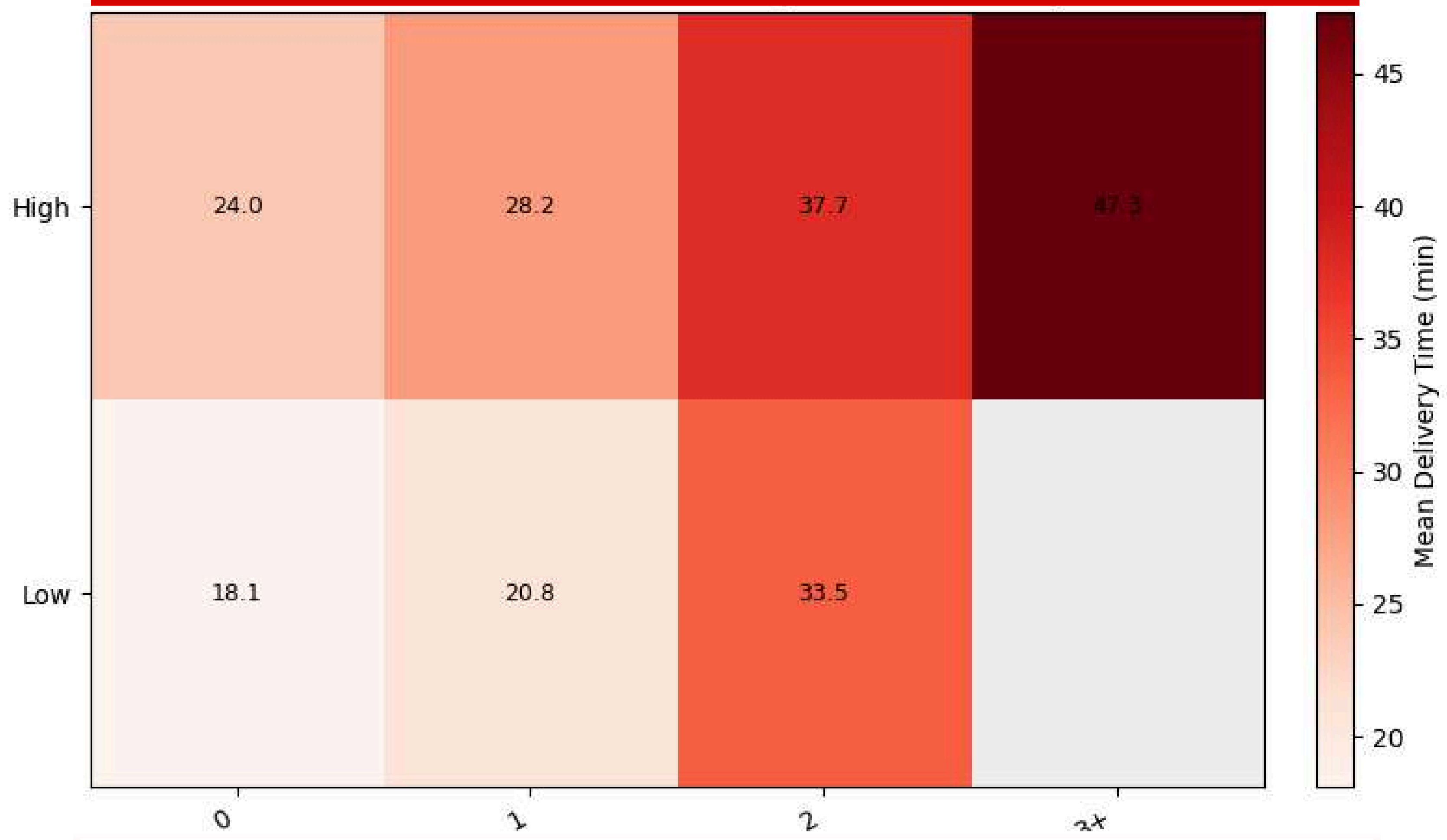
Poor vehicle condition (0) combined with extreme weather such as sandstorms, stormy, or windy conditions leads to the highest delivery times

Better vehicle conditions (1–2) reduce delivery times in all weather scenarios, though delays remain noticeable during adverse weather like sandstorms and storms

Data Analysis & Insights

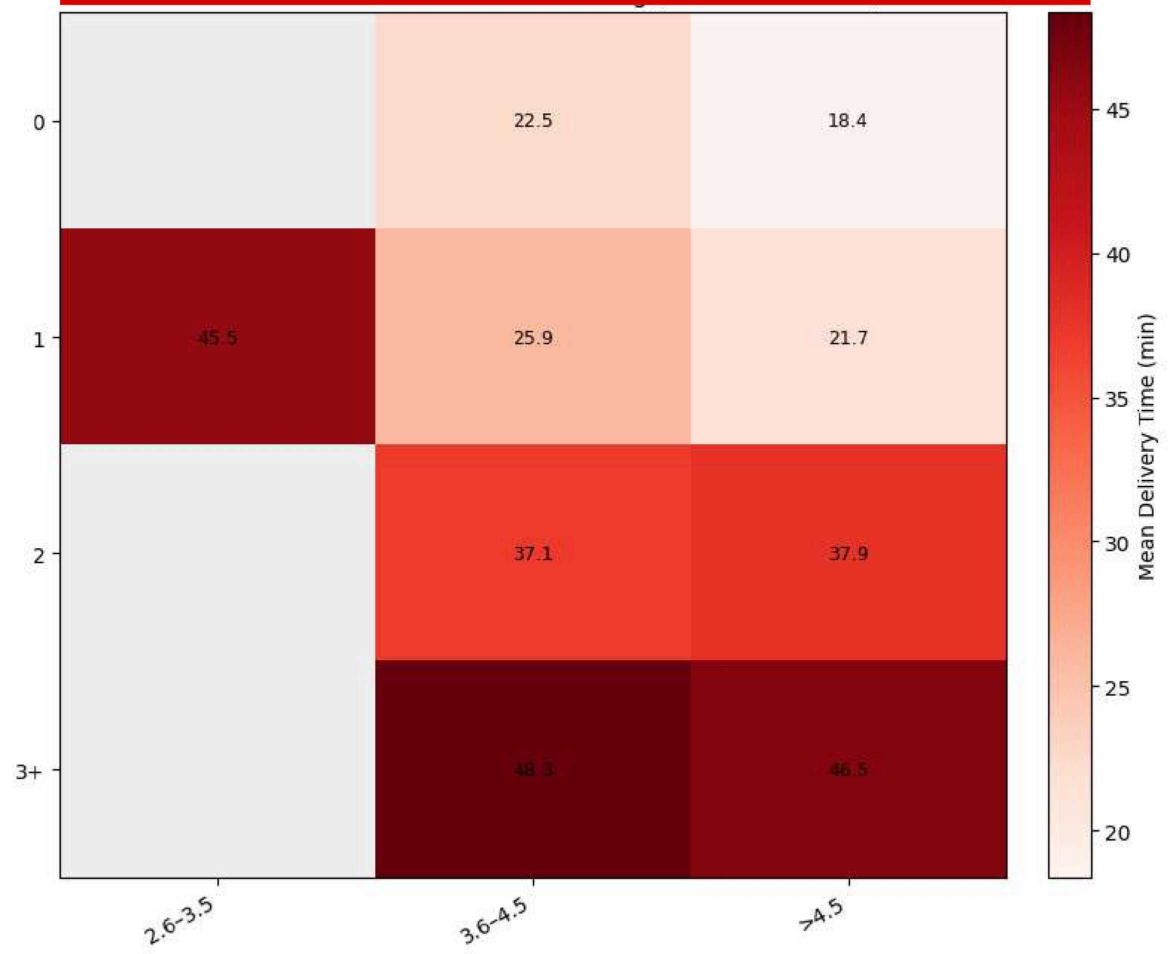
Interaction analysis

Interaction : Traffic x Workload



High traffic amplifies delays from workload—delivery time jumps from 24.0 min (0 workload) to 47.3 min (3+ workload). Even under low traffic, high workload (3+) still significantly increases delivery time (33.5 min)

Interaction : Workload x Rating Driver



Interestingly, even when a worker or service maintains a high rating, a consistently heavy workload can still result in significant delays

Delivery time spikes dramatically when workload reaches 3+ orders, regardless of driver rating. The fastest times occur at low workload (0 orders) with high ratings (18.4 min)

Dashboard

Tools

Microsoft | Power BI

This visualizations help business decision-making

Show the main delay factors

Show the main delay factors (traffic congestion, bad weather, festivals, high workload) that increase delivery time

Reveal interaction patterns

Reveal interaction patterns (e.g., congestion combined with fog results in the longest delays), providing deeper insights that help management design smarter and more efficient route planning to minimize disruptions

Compare performance across regions and conditions

Compare performance across regions and conditions (urban vs. semi-urban, driver rating, vehicle condition) so management can identify where improvements are needed

Dashboard : http://bit.ly/4fPY5K6 Streamlit : http://bit.ly/4mqXJfr

Microsoft Power BI

Recommendations & Actionable Insights

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**Focus on internal factors (workload, vehicle condition, driver rating) since they are directly controllable by the company. External factors (traffic, weather, festivals) cannot be controlled, but they still need to be anticipated by adjusting internal strategies*

Proactive Vehicle Maintenance Program

Situation	Complication
Delivery delays often stem from poor vehicle conditions (worn tires, weak brakes, or engine problems). Worse vehicle conditions increase delivery times across all weather, adding 5–7 minutes per order	Currently, Zomato has no systematic control over driver vehicle health → risk of unpredictable delays, inconsistent service quality, and higher customer complaints

Recommendation

- Mandatory light check ups every 2–3 months (partially subsidized)
- Vehicle health status reporting via app checklist at driver login
- Integration into push system scoring → drivers with "healthy" vehicles prioritized for high value orders

Partner with local workshops to provide digital service vouchers, with service data automatically feeding into the internal dashboard for monitoring and scoring

Impact			
Impact	**Reduce average delay by 5–7 mins/order (for high-risk drivers)**	**Standardize delivery quality**	**Improve customer satisfaction (CSAT) and SLA compliance over time**

Driver Performance Improvement Program

Situation	Complication
Low-rated drivers (2.6–3.5) consistently cause delivery delays, averaging ~20–25 minutes longer per order than higher-rated drivers	Currently, Zomato has only basic onboarding for new drivers, but no structured development program for underperforming low-rated drivers, causing persistent delays

Recommendation

- Training & Mentoring: Enhance driver efficiency, safety, and ability to handle external factors, ensuring stable performance under unpredictable conditions
- Incentive-penalty system → reward for measurable improvement, penalize for chronic underperformance

Pilot in selected zones with structured training, incentive rules, and rating-based allocation, then expand if delivery time variance and SLA improve

Impact			
Impact	**Delivery time for low-rated drivers expected to drop by 15–20 minutes**	**Customer satisfaction improved**	**Customer satisfaction improved (fewer complaints/refunds due to late orders)**

Recommendations & Actionable Insights

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Dynamic Order Allocation

Problem

Data shows significant delays occur when a single driver is assigned 3+ parallel orders. The current push system mostly considers proximity, ignoring other real-time variables

Recommendation

Integrate a dynamic allocation algorithm that accounts for :

Live traffic conditions (Google Maps API / internal mobility data	Flexible driver capacity limits (flexible based on conditions, not fixed)	Historical driver performance (SLA compliance, ratings, age, etc)	Vehicle condition status (updated through app or driver checklist)

Implement a priority scoring model → each order is scored and pushed to the driver with the highest compatibility score, not just the nearest one

Impact

- **Reduce 15–20% of delay cases caused by overloaded drivers**
- **Boost operational efficiency without expanding fleet size**
- **Improve customer satisfaction (faster deliveries) + driver experience (fairer allocation)**

Thank you!

Felicia Angjaya